

CaneXR: Building a Cane-Based XR Controller for Knowledge Work

Yaying Zhang*
Microsoft
Redmond, Washington, USA
yayzhang@microsoft.com

Ziming Li
Computational Media and Arts Thrust
The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, Guangdong, China
zli578@connect.hkust-gz.edu.cn

Rongkai Shi
Computational Media and Arts Thrust
The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, Guangdong, China
rongkaishi@hkust-gz.edu.cn

Brennan Jones
Faculty of Information
University of Toronto
Toronto, Ontario, Canada
brennan.jones@utoronto.ca

Hai-Ning Liang*
Computational Media and Arts Thrust
The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, Guangdong, China
hainingliang@hkust-gz.edu.cn

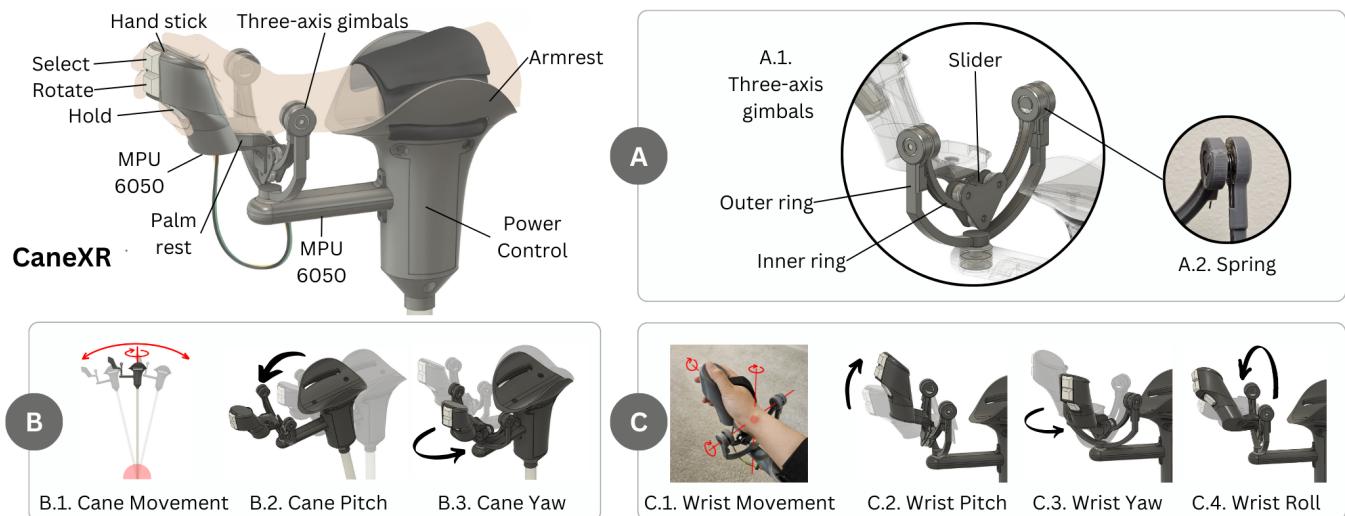


Figure 1: CaneXR: Cane-based XR controller with five degree-of-freedom (DOF) control designed for precision operations and fatigue mitigation. (A) The structure of the three-axis gimbal system with a spring solution to counterbalance gravity. (B) The cane supports 2-DOF movements. (C) The three-axis gimbal system supports 3-DOF movements with the wrist joint as the rotational center.

Abstract

While extended reality (XR) has gained traction in entertainment, its application in knowledge work remains limited. This is partially due to challenges of existing interaction methods on facilitating prolonged, high-precision operations without fatiguing the user.

*Corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720121>

Previous research suggests that a "cane" shaped design may mitigate these issues by providing ergonomic arm support. However, designs exploring this configuration are lacking. We present CaneXR, a cane-based controller with ergonomic arm support that provides controls with five degrees of freedom and operates a 3D cursor in the 3D space for object manipulation. We conducted a pilot study on its usability and received positive feedback on the adoption of support. Based on the results, we presented improvement opportunities to iterate on this prototype and expand its supporting features.

CCS Concepts

• Human-centered computing → Interaction devices; Virtual reality; Mixed / augmented reality.

Keywords

Tangible User Interface, Extended Reality, Cane Stick, Device Form Factor, Handheld Device, Ergonomic, Dynamic Arm Support, Knowledge Work

ACM Reference Format:

Yaying Zhang, Ziming Li, Rongkai Shi, Brennan Jones, and Hai-Ning Liang. 2025. CaneXR: Building a Cane-Based XR Controller for Knowledge Work. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3720121>

1 Introduction

The application of extended reality (XR) technology to knowledge work has gained increased attention due to its potential advantages as an immersive media that brings an enhanced feeling of presence, infinite display space, and improved 3D perception [6, 27, 40, 49]. Knowledge work often requires prolonged high-precision operations [56]. However, most existing XR interfaces require the user to conduct mid-air interactions with bare hands or handheld controllers. For an extended period of time, mid-air interactions can cause arm fatigue and discomfort [18, 25], an effect known as the “gorilla arm” [24], which hinders precise and comfortable input for knowledge work.

Researchers have proposed different solutions to prevent users from experiencing the gorilla arm syndrome. One solution with high potential is adopting traditional desktop input such as mouse and keyboard into XR applications [21, 22, 41, 56]. However, this approach restricts users to the desk area, partially negating the advantage of XR providing an infinite immersive space. Another potential solution is to decouple the motor space from the visual space [6], such as when operating virtual content with a tablet at hand [32] or redirecting ray casting [2]. However, this approach only alleviates the chances and effort of raising hands. The user would still need to hold the input device for prolonged use, especially in tasks requiring accuracy and precision. A new study on novel handheld devices for XR has identified the “cane” form as an unexplored but highly promising form for XR controllers [53]. This form offers arm support, potentially reducing fatigue but improving operational precision. However, a key challenge lies in balancing the need for arm support with the need to allow sufficient movement flexibility for instructing operations in XR.

In our work, we proposed a novel cane-based XR controller, **CaneXR** (see Figure 1), with ergonomic arm support that provides a five degree of freedom (DoF) control and operates a 3D cursor in the 3D space to manipulate objects. We detail the implementation of the tangible interface design and the interaction design in Section 3. Additionally, we conducted a qualitative pilot study with four XR experts and gained positive comments on CaneXR’s ergonomic supports on precision interaction and their effect on fatigue alleviation (see Section 4). We also identify areas for improvement and put forth future plans based on our findings.

2 Related Work

2.1 XR for Knowledge Work

Knowledge work is a concept first introduced by Peter Drucker [14] and refers to activities that require information workers to

apply theoretical and analytical knowledge to develop products and services. XR, as an immersive interactive media, brings several advantages to knowledge work. It enhances the sense of presence in remote collaboration [6, 13, 16, 36, 42, 44] and provides infinite display space and richer 3D visualization that facilitates data visualization and analysis [7, 9, 11, 19, 23, 34, 37, 38, 49, 54, 57]. It also strengthens 3D perception [27, 40], which is especially valuable for 3D modeling work such as product design [26, 33, 51], urban planning and architecture [49], etc. One of the challenges of adopting XR for knowledge work lies in facilitating precise and comfortable input, particularly for prolonged usage [56]. This is due to the fatigue that may arise from interacting with elevated content in an infinite 3D space via spatial, mid-air interactions [6, 52]. One solution is adopting input remapping [6], which allows the user to manipulate harder-to-reach content by alternative operations—e.g., using a handheld tablet [32] or employing a virtual pad that redirects laser cast input to higher content [2]. Yet, prolonged holding of the input device without a stable supporting platform remains an issue. Another solution is to utilize a traditional desktop mouse-keyboard setup for XR virtual content interaction [21, 22, 41, 56]. This not only facilitates high operation precision with users’ arms resting on the table, but also lowers the learning cost for users transitioning from desktop applications to XR. However, the use of these desktop accessories restricts users to the desk area, which conflicts with the central feature of XR, which is to provide an infinite immersive space with a high degree of movement.

Our work proposes a novel angle to this challenge: using a cane to provide ergonomic continuous support and mobility at the same time. In contrast to the solutions described earlier, we present a new perspective for addressing this challenge by employing a cane-based controller, and as Figure 1 shows, it offers XR users both ergonomic support and mobility simultaneously.

2.2 Ergonomic Arm Support Designs

Ergonomic arm supports can potentially minimize exertion during arm movement and alleviate arm fatigue. For instance, Tapanya et al. [47] assessed an ergonomic prototype for upper arm support during smartphone use and found that arm support can significantly reduce the activities of arm muscles, thereby reducing fatigue [47]. Ergonomic arm supports have been integrated into many daily products, especially for aiding muscle weakness [4, 29], mobility issues [15, 17], or work fatigue [8]. Such devices typically feature a curved platform that hugs the user’s forearm, potentially combined with a grip handle to move the device. For example, some walkers have employed this design to allow users to rest their arms while pushing the walker forward [15]. Our CaneXR is inspired by such a design. The type of ergonomic arm support that permits movement is called Dynamic Arm Support [12]. They employ various mechanisms including *non-actuated devices*, which need the user’s strength to drive movements [8, 30]; *passively actuated devices*, which utilize mechanisms such as counterweights, springs, or elastic bands to counter the gravitational pull on the arm [1, 28]; and *actively actuated devices* incorporate external pneumatic, electric, or hybrid energy sources to provide greater forces and more precise movement control [5, 20]. CaneXR’s design is passively actuated in that we adopted a spring solution to counterbalance gravity. Our

work distinguishes it from existing ergonomic arm support devices as an active controller. It utilizes support not simply to minimize muscle fatigue but also to leverage user movement while leaning and pivoting on the support as input to control digital data in XR applications. To our knowledge, there is no existing controller design that employs ergonomic arm support as its input mechanism for XR systems.

2.3 Cane-based Handheld Devices

Existing cane-based handheld devices are mainly designed for blind and low vision (BLV) users. One primary category is “smart canes” for BLV users to enhance their abilities to understand and navigate through their surroundings. Such design typically includes one or more sensors such as ultrasound sensors [35, 48, 50], inferred sensors [35], water sensors [3, 35], or cameras [10, 31] to detect the environment. Upon identifying noteworthy situations, they alert users via handle vibrations [10, 31, 35, 45, 46], buzzer warnings [3], or voice messages [10, 35, 50]. These smart canes function by providing signals to the user. One notable example is GesturePod, a cane that can act as an input device for white cane users and can register gestures such as tapping, twisting, twirling, and swiping to control digital devices [39]. While this work is closer to our prototype, there are some notable differences because we used the cane as an instrument to perform hand movements and control 3D cursor operations specific to XR environments (more on its features in section 3). Besides those smart canes to help detect the users’ surroundings, Microsoft Research proposed cane-based devices to simulate a virtual environment for BLV users with haptic and audio feedback [43, 55], which is the only cane-based XR controller we have noticed. However, all of these existing cane-based controllers function as detectors rather than supporters. The user is required to lift the cane in order to use them. In contrast, our work aims to allow the user to rest on the cane on the floor and perform precise and extended operations over long periods of time. To the best of our knowledge, we have not come across a handheld cane-based device that serves this specific dual purpose of providing continuous resting support and enabling interactions with virtual objects.

3 CaneXR Prototype Design

We propose CaneXR, a cane-based controller that provides 5-DoF movement for the user to control a 3D cursor to manipulate virtual objects in XR environments. We utilized Fusion 360 to create a 3D model of the prototype, followed by 3D printing the components using PLA plastic. The various parts were then assembled using mainly screws. Please refer to the supplementary video included with this paper to see how the prototype and the interactions with it work.

3.1 Tangible Interface Design: Five-DoF Dynamic Support

In this section, we detail the hardware design of the CaneXR controller (see Figure 1). The core structure of CaneXR draws inspiration from established ergonomic support devices, such as forearm crutches and arm supports described in the previous section. The physical design of the CaneXR comprises three main components: 1) a cane stick equipped with an armrest, 2) a three-axis gimbal

system facilitating wrist movement, and 3) a hand stick with a palm rest.

3.1.1 Cane Stick with Arm Rest. The cane stick serves as the primary support for the user’s arm, providing stability from the ground up. At the top of the cane stick, an armrest platform allows the user to rest their arm comfortably. A strap can be utilized to securely bind the user’s arm to the cane, enabling them to lift the cane with one hand. While supporting the weight of the user’s arm, the cane stick offers two degrees of freedom: forward-backward movement (pitch) and left-right rotation (yaw) (see Figure 1-B). Although users can theoretically conduct roll rotation by moving their arms closer to or farther from their torso (abduction/adduction of the shoulder joint)—akin to the motion of a “chicken dance”—this movement is excluded due to its limited range and reduced comfort compared to the primary motions. Beneath the armrest, a chamber houses the power control inside, with a case for a 5-volt 18650 rechargeable battery. The power switch and indicator light are located on the exterior of the chamber. Near the top of the cane stick, a horizontal bar extends outward, to which an inertial measurement unit (IMU) sensor, MPU6050, is affixed underneath to monitor arm rotation. This structure connects to the three-axis gimbal system.

3.1.2 Three-Axis Gimbal System. A gimbal is a pivoted support that permits the rotation of an object about an axis. The three-axis gimbal system consists of an outer ring, an inner ring, and a slider on the inner ring (Figure 1-A). It provides three DOF for movement in pitch, yaw, and roll (Figure 1-C), with the user’s wrist joint as the rotational center (Figure 1-C.1). The outer ring has a pin at the bottom that connects to the horizontal bar through a 608 bearing, facilitating yaw rotation. The outer ring also includes two 607 bearings at each end, which connect to the two 607 bearings on each end of the inner ring to enable pitch rotation. The inner ring, shaped like a half circle, has grooves on both its inner and outer sides, creating a rail for the slider to traverse in the roll direction. The slider contains three 605 bearings, with two moving along the inner groove of the ring and one along the outer groove. Two triangular shells at the front and back secure the three bearings in place with three pins. The slider incorporates an L-shaped connector to attach to the palm rest of the hand stick.

One challenge we faced was that the weight center of the hand was not at the joint center of the wrist. Consequently, gravity tends to pull the user’s hand downward in the pitch direction, resulting in increased effort to keep the hand straight. There are multiple potential solutions, including modifying the size and weight center of the hand stick or the gimbals to counterbalance the weight of the hand, or involving vector control motors to regulate torque on the gimbals precisely. Ultimately, we adopted a simple spring solution to counteract the effects of gravity (Figure 1-A.2). On each end of the gimbal rings, where the inner ring connects to the outer ring, we installed a 60° torsion spring to apply outward pressure between the two rings. We selected the specifications of the springs to provide approximately 400 grams of lift—the average weight of an adult’s hand—when the springs are compressed to a 0° position. This approach effectively allows the user’s hand to rest naturally in an upright position.

3.1.3 Hand Stick with Palm Rest. The hand stick features a palm rest at its base, which connects to the slider of the gimbal system, allowing the user to rest their hand while gripping the hand stick. We let the stick tilt forward at an angle of 24° to enhance user comfort (Figure 1). The chosen tilting degree was the most comfortable one after trying multiple 3D-printed prototypes. The stick was equipped with three buttons: two smaller buttons designed to be pressed by the user's index and middle fingers and a larger button positioned below to be pressed by the ring and pinky fingers, as can be seen from Figures 1 and 2. From top to bottom, we refer to them as the Select, Rotate, and Hold buttons and will explain their use in Section 3.2 soon. To facilitate movement of the hand stick without the risk of accidentally pressing the buttons, a strap securely binds the stick to the user's palm. A second MPU6050 IMU sensor is mounted at the bottom of the hand stick to detect wrist rotation. Inside the hand stick, an ESP-WROOM-32 microcontroller module is integrated, responsible for transmitting the status of the three buttons and the two IMU sensors to the XR application via WiFi using User Datagram Protocol (UDP).

3.2 Interaction Design: 3D Cursor

3.2.1 3D Cursor. CaneXR controller possesses the advantage of allowing the user to conduct micro-movements while resting their hand. As a result, common XR interactions such as a 1:1 mapped virtual hand or laser pointing are not suitable due to their requirement of significant hand movements. To meet our needs, we developed a 3D cursor interaction extending from the traditional mouse-controlled 2D cursor system in desktop applications. The 3D cursor, represented as a floating arrowhead in the virtual space, is maneuvered through the user's elbow and wrist movement on the CaneXR controller. Besides allowing micro-movements for control, the 3D cursor also possesses three benefits. First, it effectively navigates complex 3D scenes where objects may overlap or obscure one another, a capability laser casting lacks. Second, its sharp tip provides superior precision for object selection compared to virtual hands. Finally, the 3D cursor's design aligns with familiar desktop software solutions, making it more intuitive for knowledge workers.

3.2.2 Cursor Movement and Coordination System. We implemented a clutch mechanism to decouple hand and cursor movements, akin to a 2D mouse cursor—it follows the hand movement when the mouse is held against the table and stays still when the user releases or lifts the mouse. CaneXR users can only move our 3D cursor when “holding” the hand stick tighter, i.e., by pressing down the **Hold button**. The cursor remains stationary when the user disengages the Hold button, thus allowing the user to adjust their hand pose without affecting the cursor's position. Once pressing and holding the Hold button, the user can manipulate the cursor position through specific hand movements (see Figure 2-A). The user can swing their arm or hand left or right on the yaw axis to adjust the cursor's x position. The hand rotation has a smaller control-display ratio (CD ratio) than the arm for more precise operations. The user can combine both arm and hand moves for compound adjustments. For modifying the cursor's y position, the user can tilt the hand up or down on the pitch axis. Similarly, the user can tilt the cane forward or backward to alter the cursor's z position.

Since the user can locomote in the 3D space, the cursor's moving directions adapt to the user's orientation. Each time when the Hold button is engaged, the cursor's coordination system recalibrates: the forward direction aligns to the direction where the user's center eye direction is projected to the horizontal plane. The up direction remains vertical, while the right direction is perpendicular to the forward and up directions.

3.2.3 Object Manipulation. We designed our interaction to fulfill common CAD tasks of hovering, selecting, moving, and rotating. When the cursor's tip contacts a manipulatable object, the object enters into a “hover” state, indicating its readiness for selection. Then the user can press the Select button to select it or move the cursor away to cancel the hovering (Figure 2-B). Once an object is selected, the user can either dismiss the selection by clicking on an empty space or manipulate the selected object's position and rotation in two ways. The first way is **gizmo manipulation**. Upon selecting an object, a gizmo appears, featuring three positional handles and three rotational handles (Figure 2-B). The user can drag these handles (by hovering on the handle and holding the Select button while moving the cursor) (Figure 2-D) to translate or rotate in the corresponding direction. The second method, **free manipulation** (Figure 2-C), triggered by the user hovering on the selected object and then holding either the Select or Rotate button, allows the user to directly translate or rotate the object. The user can hold the Select button while moving the cursor to let the object follow the cursor's position. Alternatively, they can hold the Rotate button while rotating their wrist. When rotating, the cursor movement is paused, and the user's wrist rotation delta will be applied to the object.

4 Pilot Study

To preliminarily understand the usability of our prototype, we conducted a qualitative pilot study with four participants (P0 - P3, 3m/1f, age 25 - 27, all experienced with XR technologies and HCI research).

The pilot study lasted for approximately 30 minutes for each participant. In the first 15 minutes, we allowed participants to engage in 2D CAD operations using a mouse and keyboard, letting them recognize the limitations of the 2D interface in terms of 3D modeling, specifically the absence of depth perception. Then, we let the participant experience the same CAD operations in VR with a Quest Pro headset and its associated controllers. Subsequently, we explained the fatigue issue caused by long usage and presented the CaneXR controller to the participants. Unfortunately, due to sensor issues at the time of the study, we were unable to provide real-time testing of the interaction. Instead, we played a recorded demonstration video displaying both the user's movement and the view in VR (as attached in the supplementary material) that showcased manipulating the cursor and objects with the CaneXR controller. While watching, the participant shadowed the movement with the CaneXR controller as shown in the video, engaging in a “Wizard-of-Oz” scenario. With this approach, the participants were able to sense the interaction.

For the second half of the study, we interviewed participants about their opinions on the CaneXR prototype. Our questions focused on two aspects: the tangible interface design (hardware) and

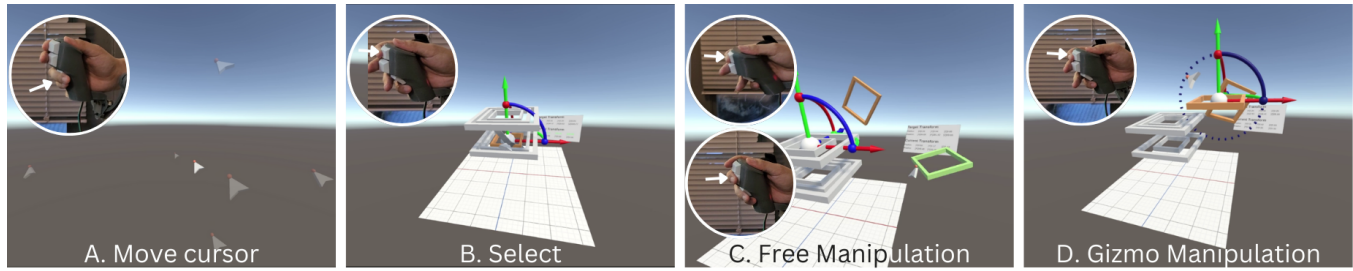


Figure 2: CaneXR Interactions in an XR environment: (A) moving the 3D cursor with the Hold button, (B) selecting an object, (C) manipulating an object in 6-DOF (free manipulation), and (D) manipulating an object in 1-DOF (gizmo manipulation).

the interaction design (software). Regarding the tangible interface design, we first asked the participants to provide their overall feedback while holding and moving with the controller. Then, we asked their opinions on the armrest, the gimbal system, and the hand stick, respectively. For the interaction design, we again asked for overall comments first, followed by specific inquiries on the cursor moving and the object manipulation. The experimenter asked follow-up questions when needed. When an interesting discussion appeared, we sought opinions from subsequent participants.

To analyze the data, we first did an open coding process across the recordings sentence by sentence and marked the key opinions for each sentence. We then conducted axial coding by grouping participants' opinions into three themes (support, fatigue, and precision) and six aspects (button, hand stick, wrist, elbow, straps, and interaction). We were especially interested in these themes and aspects to further iterate the prototype. Within each aspect and theme, we compared and concluded participants' opinions.

5 Results and Discussion

5.1 Support may reduce fatigue with some constrained feeling

Providing support is the key objective of our CaneXR design. We gladly observed that participants mainly hold positive opinions on the support design. In terms of fatigue, most participants agreed that the support design could reduce fatigue. P0 mentioned "[CaneXR] solved the sore arm problem. It's indeed hard to feel fatigued if I put my arm and hand on it." P1 stated, "CaneXR provided a point of support, which saves effort in long-term use." P2 commented, "[With CaneXR] I felt less burden because there's support." However, participants expressed this support somewhat constrains their movement. P1 thought CaneXR's support and strap design were somewhat troublesome, and limited his movements to some degree. P2 expressed that CaneXR gave more "feeling of constraints." P3 even found the arm restriction uncomfortable, preventing her from prolonged use: "[The arm support is] too restricting. I'd like to move freely and get my arm straight when I want. This [design] made my [elbow] joint hurt."

5.2 Support may increase precision, but movement mapping requires improvement

In terms of precision, most participants predict that the CaneXR interface might offer improved precision due to the added support.

P1 remarked, "When I play VR shooting games, I've always wished to add support to the controller. So having something to rest on would definitely help with high-precision operations." P3 provided another example regarding the application of eyeliner, where it can be challenging to achieve the desired contouring on the eyelids. P3 mentioned that experienced people often rest their hands against their cheekbones for stability during this process. However, P3 felt hand support would be sufficient for precise operations, and the elbow support was unnecessary. P2 also felt CaneXR could provide better precision: "The precision winner must be the CaneXR [comparing to Quest controller]." Moreover, he noted that the passive haptic feedback that the support increased his awareness of his hand position: "The support has an 'anchored' feeling? Yea, I can also better determine the position of my hand."

Designing an effective control-display mapping could be crucial for achieving operational precision. One challenge in our interaction is that we have to map the *rotation* of the arm/hand to the *position* of the cursor, which may not be as intuitive as other control methods, such as a desktop mouse, where the hand's x-y position directly maps to the cursor's x-y position, or the Quest controller, where the virtual controller tracks the real controller with exactitude. For the pilot study, we implemented a 1-degree-to-1-centimeter mapping ratio for cursor control, whereby each degree of axis rotation from hand or arm input triggers a cursor movement of one-centimeter distance in the corresponding direction. With this ratio, P0 felt the cursor in CaneXR moved faster than anticipated, whereas the Quest controller behaved more closely with user expectations. We could adjust the cursor to move slower, but it would be harder for the user to control the cursor for long distances. To resolve this, we could apply a dynamic mapping ratio in future iterations to make the cursor move slower when the user's hand moves slower, and vice versa—a trick desktop mouse does.

5.3 Challenges for button design

P2 and P3 felt the button positions were not comfortable enough to use, specifically noting that they had to exert extra effort to move their index and middle fingers upward to reach them. In contrast, P0 and P1 did not raise this concern. We felt it could be a challenge to accommodate varying hand sizes. Users with smaller hands might need to lift their fingers to reach the buttons. Conversely, lowering the button positions could prompt users with larger hands to lower their fingers, potentially causing them to squeeze their palms against the rest. P0 also expressed concerns about the Hold

button being operated by the least frequently used fingers (the ring and pinky fingers), potentially leading to discomfort. Another refinement opportunity is the choice of button types. Our prototype currently features blue mechanical key switches, known for their clicky sound and crisp haptic feedback, but require more actuation force. However, P1, P2, and P3 all preferred a linear switch over the clicky switch. P1 said, *“I don’t like to use it [the clicky keys] for a long time.”* Compared to those on conventional keyboards, our buttons are harder to press due to the vertical orientation of our controller, which prevents users from leveraging gravity when pressing keys. Besides, P3 suggested making the button smooth and making the pressure detectable: *“I can micro-control the degree of pressing - pressing lightly or hard... When I press a little, it moves little; When I press hard, it moves big.”* Based on this feedback, we might consider switching to lighter, linear key switches like red switches, except for the Hold button. P1 and P2 felt the Hold button could remain clicky. P1 said, *“For game controllers’ [trigger] buttons, I usually don’t want them to be clicky. But we are making it behave like a mouse, so I hope the Hold button has a clicky feeling because their function is different.”* P2 said, *“Because the Hold button needs to give the feedback of zero or one—I need to know whether I pressed it.”* According to this feedback, we consider redesigning the Hold button to use the thumb with it instead of the ring and pinky finger. For the button type, we consider using a linear button for the Select and Rotate buttons and keeping a clicky button for the Hold button.

5.4 Reconsideration of resting hand pose

For our prototype, the resting hand is positioned vertically, with the palm facing left and the ulnar side facing down. P0 felt that moving the cursor up and down felt uncomfortable because it required the hand’s radial and ulnar deviation, which is not as comfortable as flexion and extension. He suggested making it horizontal, where the palm faces down, and the ulnar side faces right. We sought opinions from other participants but received varied feedback. P1 favored the horizontal position because *“it felt more like holding a mouse.”* But P2 felt horizontal is *“more comfortable because the palm rest is down there.”* P3 initially preferred vertical because *“[horizontal] felt weird,”* but she also acknowledged the horizontal pose’s advantage of resembling a mouse. For our future iterations, we will explore the horizontal resting hand pose and compare it against the current vertical pose.

6 Limitations and Future Work

This work has two limitations. First, we only asked a limited number of participants with past XR and HCI experiences to provide feedback on CaneXR in the pilot study. Although this is cost-effective for collecting professional opinions, we plan to extend the study further by inviting more participants with different backgrounds to share their feedback. Second, the pilot study followed the Wizard-of-Oz approach to receive preliminary feedback on CaneXR, rather than having participants experience using the prototype for real object manipulation. In the future, we plan to implement and iterate functional prototypes based on the pilot results. We will conduct lab experiments to examine the interface’s effect on fatigue and precision for XR operations in knowledge work and compare it with controller- and mouse-based interactions.

7 Conclusion

This paper presented CaneXR, a novel cane-based controller that provides arm support and five DOFs for arm movement to control a 3D cursor in XR. We detailed the implementation of the tangible interface design and the interaction design. Our pilot study showed the potential of the arm support design to reduce fatigue and increase precision, but it had some limitations, such as a sense of constrained and unexpected movement mapping. We hope CaneXR can inspire HCI and XR researchers to prototype cane-based interfaces and design novel interactions to support knowledge work in the future.

References

- [1] Active Mobility Systems. [n.d.]. Multilink Arm Support | Active Mobility Systems. <https://www.activemobility.com.au/multilink-dynamic-arm-support>. Accessed: 2025-1-22.
- [2] Carlos Andujar and Ferran Argelaguet. 2007. Virtual Pads: Decoupling Motor Space and Visual Space for Flexible Manipulation of 2D Windows within VEs. In *2007 IEEE Symposium on 3D User Interfaces*. <https://doi.org/10.1109/3DUI.2007.340781>
- [3] Asraful Islam Apu, Al-Akhir Nayan, Jannatul Ferdous, and Muhammad Golam Kibria. 2022. IoT-Based Smart Blind Stick. In *Proceedings of the International Conference on Big Data, IoT, and Machine Learning*, Mohammad Shamsul Arefin, M. Shamim Kaiser, Anirban Bandyopadhyay, Md. Atiqur Rahman Ahad, and Kanad Ray (Eds.). Springer Singapore, Singapore, 447–460.
- [4] Armon Products. [n.d.]. Edero Assistive — Armon Products. <https://armonproducts.com/eder-assistive/>. Accessed: 2025-1-22.
- [5] Armon Products. [n.d.]. Salvum — Armon Products. <https://armonproducts.com/salvum/>. Accessed: 2025-1-22.
- [6] Verena Biener, Eyal Ofek, Michel Pahud, Per Ola Kristensson, and Jens Grubert. 2023. *Extended Reality for Knowledge Work in Everyday Environments*. Springer International Publishing, Cham, 21–56. https://doi.org/10.1007/978-3-031-05804-2_2
- [7] Verena Biener, Daniel Schneider, Travis Gesslein, Alexander Otte, Bastian Kuth, Per Ola Kristensson, Eyal Ofek, Michel Pahud, and Jens Grubert. 2020. Breaking the Screen: Interaction Across Touchscreen Boundaries in Virtual Reality for Mobile Knowledge Workers. *IEEE Transactions on Visualization and Computer Graphics* 26, 12 (2020), 3490–3502. <https://doi.org/10.1109/TVCG.2020.3023567>
- [8] BONTEC. [n.d.]. BONTEC Ergonomic Arm Rest Support for Desk Armrest Pad Rotating Wrist. <https://bontec.co.uk/products/bontec-ergonomic-arm-rest-support-for-desk-armrest-pad-rotating-wrist-rest-holder>. Accessed: 2025-1-22.
- [9] Tone Bratteteig and Ina Wagner. 2012. Spaces for participatory creativity. *CoDesign* 8, 2-3 (2012), 105–126. <https://doi.org/10.1080/15710882.2012.672576> arXiv:https://doi.org/10.1080/15710882.2012.672576
- [10] Qingtian Chen, Muhammad Khan, Christina Tsangouri, Christopher Yang, Bing Li, Jizhong Xiao, and Zhigang Zhu. 2017. CCNY Smart Cane. In *2017 IEEE 7th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*. 1246–1251. <https://doi.org/10.1109/CYBER.2017.8446303>
- [11] Andreas Dengel, Stefan Agne, Bertin Klein, Achim Ebert, and Matthias Deller. 2006. Human-centered interaction with documents. In *Proceedings of the 1st ACM International Workshop on Human-Centered Multimedia* (Santa Barbara, California, USA) (*HCM '06*). Association for Computing Machinery, New York, NY, USA, 35–44. <https://doi.org/10.1145/1178745.1178755>
- [12] Loek A Van der Heide, Bob van Nijhuijs, Arjen Bergsma, Gert Jan Gelderblom, Dick J van der Pijl, and Luc P de Witte. 2014. An overview and categorization of dynamic arm supports for people with decreased arm function. *Prosthetics and Orthotics International* 38, 4 (2014), 287–302. <https://doi.org/10.1177/0309364613498538> arXiv:https://doi.org/10.1177/0309364613498538 PMID: 23950551.
- [13] James Dominic, Brock Tubre, Charles Ritter, Jada Houser, Colton Smith, and Paige Rodeghero. 2020. Remote Pair Programming in Virtual Reality. In *2020 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. 406–417. <https://doi.org/10.1109/ICSME46990.2020.00046>
- [14] Peter Drucker. 1959. *The Landmarks of Tomorrow*. Harper & Brothers, New York, NY, US.
- [15] ELENKER. [n.d.]. ELENKER Upright Walker | Stand Up Folding Rollator With Seat — Elenker. <https://www.elenkerwalker.com/collections/upright-walker>. Accessed: 2025-1-22.
- [16] Anthony Elliott, Brian Peiris, and Chris Parnin. 2015. Virtual Reality in Software Engineering: Affordances, Applications, and Challenges. In *2015 IEEE/ACM 37th IEEE International Conference on Software Engineering*, Vol. 2. 547–550. <https://doi.org/10.1109/ICSE.2015.7455000>

- //doi.org/10.1109/ICSE.2015.191
- [17] Ergoactives. [n.d.]. Ergobaum Junior Forearm Crutches (Pair) — Ergoactives. <https://www.ergoactives.com/collections/ergobaum/products/ergobaum-junior>. Accessed: 2025-1-22.
 - [18] João Marcelo Evangelista Belo, Anna Maria Feit, Tiare Feuchtnner, and Kaj Grønbaek. 2021. XRgonomics: facilitating the creation of ergonomic 3D interfaces. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–11.
 - [19] Travis Gesslein, Verena Biener, Philipp Gagel, Daniel Schneider, Per Ola Kristensson, Eyal Ofek, Michel Pahud, and Jens Grubert. 2020. Pen-based Interaction with Spreadsheets in Mobile Virtual Reality. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 361–373. <https://doi.org/10.1109/ISMAR50242.2020.00063>
 - [20] R. A. R. C. Gopura, Kazuo Kiguchi, and Yang Li. 2009. SUEFUL-7: A 7DOF upper-limb exoskeleton robot with muscle-model-oriented EMG-based control. In *2009 IEEE/RSJ International Conference on Intelligent Robots and Systems*. 1126–1131. <https://doi.org/10.1109/IROS.2009.5353935>
 - [21] Jens Grubert, Eyal Ofek, Michel Pahud, and Per Ola Kristensson. 2020. Back to the future: Revisiting mouse and keyboard interaction for hmd-based immersive analytics. *arXiv preprint arXiv:2009.02927* (2020).
 - [22] Jens Grubert, Lukas Witzani, Eyal Ofek, Michel Pahud, Matthias Kranz, and Per Ola Kristensson. 2018. Text Entry in Immersive Head-Mounted Display-Based Virtual Reality Using Standard Keyboards. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 159–166. <https://doi.org/10.1109/VR.2018.8446059>
 - [23] Leman Figen Gül. 2018. Studying gesture-based interaction on a mobile augmented reality application for co-design activity. *Journal on Multimodal User Interfaces* 12 (2018), 109–124. <https://doi.org/10.1007/s12193-017-0252-0>
 - [24] Jeffrey T. Hansberger, Chao Peng, Shannon L. Mathis, Vaidyanath Areyur Shanthakumar, Sarah C. Meacham, Lizhou Cao, and Victoria R. Blakely. 2017. Dispelling the Gorilla Arm Syndrome: The Viability of Prolonged Gesture Interactions. In *Virtual, Augmented and Mixed Reality*, Stephanie Lackey and Jessie Chen (Eds.). Springer International Publishing, Cham, 505–520.
 - [25] Juan David Hincapié-Ramos, Xiang Guo, Paymahn Moghadasian, and Pourang Irani. 2014. Consumed endurance: a metric to quantify arm fatigue of mid-air interactions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 1063–1072.
 - [26] Nikola Horvat, Steffen Kunnen, Mario Štorga, Arun Nagarajah, and Stanko Škec. 2022. Immersive virtual reality applications for design reviews: Systematic literature review and classification scheme for functionalities. *Advanced Engineering Informatics* 54 (2022), 101760. <https://doi.org/10.1016/j.aei.2022.101760>
 - [27] Nikola Horvat, Stanko Škec, Tomislav Martinec, Fanika Lukačević, and Marija Majda Perišić. 2019. Comparing Virtual Reality and Desktop Interface for Reviewing 3D CAD Models. *Proceedings of the Design Society: International Conference on Engineering Design* 1, 1 (2019), 1923–1932. <https://doi.org/10.1017/dsi.2019.198>
 - [28] JAECO Orthopedic. [n.d.]. JAECO WREX | JAECO Orthopedic. <https://jaecoorthopedic.com/product/jaeco-wrex/>. Accessed: 2025-1-22.
 - [29] JAECO Orthopedic. [n.d.]. MultiLink Arm | JAECO Orthopedic. <https://jaecoorthopedic.com/product/multilink-arm/>. Accessed: 2025-1-22.
 - [30] JAECO Orthopedic. [n.d.]. Standard Mobile Arm Support | JAECO Orthopedic. <https://jaecoorthopedic.com/product/standard-mobile-arm-support/>. Accessed: 2025-1-22.
 - [31] Yongsik Jin, Jonghong Kim, Bumhwi Kim, Rammohan Mallipeddi, and Minho Lee. 2015. Smart Cane: Face Recognition System for Blind. In *Proceedings of the 3rd International Conference on Human-Agent Interaction* (Daegu, Kyungpook, Republic of Korea) (HAI '15). Association for Computing Machinery, New York, NY, USA, 145–148. <https://doi.org/10.1145/2814940.2814952>
 - [32] Khanh-Duy Le, Tanh Quang Tran, Karol Chlasta, Krzysztof Krejtz, Morten Fjeld, and Andreas Kunz. 2021. VXSlate: Exploring Combination of Head Movements and Mobile Touch for Large Virtual Display Interaction. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference* (Virtual Event, USA) (DIS '21). Association for Computing Machinery, New York, NY, USA, 283–297. <https://doi.org/10.1145/3461778.3462076>
 - [33] Rui Liu, Chao Peng, Yunbo Zhang, Hannah Husarek, and Qi Yu. 2021. A survey of immersive technologies and applications for industrial product development. *Computers & Graphics* 100 (2021), 137–151.
 - [34] Laura Maftai and Chris Harty. 2015. Designing in caves: using immersive visualisations in design practice. *International Journal of Architectural Research: ArchNet-IJAR* 9, 3 (2015), 53–75. <https://doi.org/10.26687/archnet-ijar.v9i3.693>
 - [35] Ayat Nada, Samia Mashelly, Mahmoud A Fakhr, and Ahmed F Seddik. 2015. Effective fast response smart stick for blind people. In *Proceedings of the second international Conference on Advances in bio-informatics and environmental engineering-ICABEE*.
 - [36] Cuong Nguyen, Stephen DiVerdi, Aaron Hertzmann, and Feng Liu. 2017. CollaVR: Collaborative In-Headset Review for VR Video. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology* (Québec City, QC, Canada) (UIST '17). Association for Computing Machinery, New York, NY, USA, 267–277. <https://doi.org/10.1145/3126594.3126659>
 - [37] Erwan Normand and Michael J. McGuffin. 2018. Enlarging a Smartphone with AR to Create a Handheld VESAD (Virtually Extended Screen-Aligned Display). In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 123–133. <https://doi.org/10.1109/ISMAR.2018.00043>
 - [38] Eoin O'Brien, Baptiste Jacquouton, Antoine Moineau, and Abraham G. Campbell. 2019. Wikipedia in Virtual Reality and HowText-based Media can be Explored in Virtual Reality. In *Proceedings of the 2019 International Conference on Artificial Intelligence and Advanced Manufacturing* (Dublin, Ireland) (AIAM 2019). Association for Computing Machinery, New York, NY, USA, Article 70, 9 pages. <https://doi.org/10.1145/3358331.3358401>
 - [39] Shishir G. Patil, Don Kurian Dennis, Chirag Pabbaraju, Nadeem Shaheer, Harsha Vardhan Simhadri, Vivek Seshadri, Manik Varma, and Prateek Jain. 2019. GesturePod: Enabling On-device Gesture-based Interaction for White Cane Users. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 403–415. <https://doi.org/10.1145/3332165.3347881>
 - [40] Bónsa Regassa Hunde and Abraham Debebe Woldeyohannes. 2022. Future prospects of computer-aided design (CAD) – A review from the perspective of artificial intelligence (AI), extended reality, and 3D printing. *Results in Engineering* 14 (2022), 100478. <https://doi.org/10.1016/j.rineng.2022.100478>
 - [41] Daniel Schneider, Alexander Otte, Travis Gesslein, Philipp Gagel, Bastian Kuth, Mohamad Shahm Damlakhi, Oliver Dietz, Eyal Ofek, Michel Pahud, Per Ola Kristensson, Jörg Müller, and Jens Grubert. 2019. ReconVigRation: Reconfiguring Physical Keyboards in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics* 25, 11 (2019), 3190–3201. <https://doi.org/10.1109/TVCG.2019.2932239>
 - [42] Geetika Sharma, Gautam Shroff, and Prasun Dewan. 2011. Workplace collaboration in a 3D Virtual Office. In *2011 IEEE International Symposium on VR Innovation*. 3–10. <https://doi.org/10.1109/ISVR.2011.5759582>
 - [43] Alexa F. Siu, Mike Sinclair, Robert Kovacs, Eyal Ofek, Christian Holz, and Edward Cutrell. 2020. Virtual Reality Without Vision: A Haptic and Auditory White Cane to Navigate Complex Virtual Worlds. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376353>
 - [44] Spatial Systems, Inc. 2025. Spatial - Create Immersive UGC, Virtual Classrooms, Experimental Marketing. <https://spatial.io/>. Accessed: 2025-1-22.
 - [45] Takeshi Tanabe, Yasuhiro Fujimoto, Kiyohiko Nunokawa, Kouki Doi, and Shuichi Ino. 2023. White Cane-Type Holdable Device Using Illusory Pulling Cues for Orientation & Mobility Training. *IEEE Access* 11 (2023), 28706–28714. <https://doi.org/10.1109/ACCESS.2023.3259965>
 - [46] Takeshi Tanabe, Kiyohiko Nunokawa, Kouki Doi, and Shuichi Ino. 2022. Training System for White Cane Technique Using Illusory Pulling Cues Induced by Asymmetric Vibrations. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 30 (2022), 305–313. <https://doi.org/10.1109/TNSRE.2022.3148770>
 - [47] Weerasak Tapanya, Rungthip Puntumetakul, Manida Swangnetr Neubert, Torkamol Humsawong, and Rose Boucatt. 2021. Ergonomic arm support prototype device for smartphone users reduces neck and shoulder musculoskeletal loading and fatigue. *Applied Ergonomics* 95 (2021), 103458. <https://doi.org/10.1016/j.apergo.2021.103458>
 - [48] I. Ulrich and J. Borenstein. 2001. The GuideCane-applying mobile robot technologies to assist the visually impaired. *IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans* 31, 2 (2001), 131–136. <https://doi.org/10.1109/3468.911370>
 - [49] Minna Vasarainen, Sami Paavola, and Liubov Vetoshkina. 2021. A Systematic Literature Review on Extended Reality: Virtual, Augmented and Mixed Reality in Working Life. *International Journal of Virtual Reality* 21, 2 (Oct. 2021), 1–28. <https://doi.org/10.20870/IJVR.2021.21.2.4620>
 - [50] Mohd Helmy Abd Wahab, Amirul A. Talib, Herdawatie A. Kadir, Ayob Johari, Ahmad Noraziah, Roslina M. Sidek, and Ariffin A. Mutalib. 2011. Smart cane: Assistive cane for visually-impaired people. *arXiv preprint arXiv:1110.5156* (2011).
 - [51] Josef Wolfartsberger. 2019. Analyzing the potential of Virtual Reality for engineering design review. *Automation in Construction* 104 (2019), 27–37. <https://doi.org/10.1016/j.autcon.2019.03.018>
 - [52] Difeng Yu, Xueshi Lu, Rongkai Shi, Hai-Ning Liang, Tilman Dingler, Eduardo Velloso, and Jorge Goncalves. 2021. Gaze-Supported 3D Object Manipulation in Virtual Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 734, 13 pages. <https://doi.org/10.1145/3411764.3445343>
 - [53] Yaying Zhang, Rongkai Shi, and Hai-Ning Liang. 2024. Stick-To-XR: Understanding Stick-Based User Interface Design for Extended Reality. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (DIS '24). Association for Computing Machinery, New York, NY, USA, 168–179. <https://doi.org/10.1145/3643834.3661627>
 - [54] Lixiang Zhao, Tobias Isenberg, Fuqi Xie, Hai-Ning Liang, and Lingyun Yu. 2025. SpatialTouch: Exploring Spatial Data Visualizations in Cross-Reality.

- IEEE Transactions on Visualization and Computer Graphics* 31, 1 (2025), 897–907. <https://doi.org/10.1109/TVCG.2024.3456368>
- [55] Yuhang Zhao, Cynthia L. Bennett, Hrvoje Benko, Edward Cutrell, Christian Holz, Meredith Ringel Morris, and Mike Sinclair. 2018. Enabling People with Visual Impairments to Navigate Virtual Reality with a Haptic and Auditory Cane Simulation. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (*CHI '18*). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3173574.3173690>
- [56] Qian Zhou, George Fitzmaurice, and Fraser Anderson. 2022. In-Depth Mouse: Integrating Desktop Mouse into Virtual Reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (*CHI '22*). Association for Computing Machinery, New York, NY, USA, Article 354, 17 pages. <https://doi.org/10.1145/3491102.3501884>
- [57] Daniel Zielasko, Marcel Krüger, Benjamin Weyers, and Torsten W. Kuhlen. 2019. Passive Haptic Menus for Desk-Based and HMD-Projected Virtual Reality. In *2019 IEEE 5th Workshop on Everyday Virtual Reality (WEVR)*. 1–6. <https://doi.org/10.1109/WEVR.2019.8809589>